

# Medical Diagnosis and Report Generation for Endoscopic Procedures Using Vision-Language Models

Praval Pattam, Theenesh Potluri, Pranav Sai Sarvepalli

Under the Guidance of: Dr. Jayaraj P B, Srinivasa T M

National Institute of Technology Calicut

B.Tech CSE Semester 8 Final Evaluation

# Agenda

- 1 Introduction
- 2 Methodology
- 3 Architecture
- 4 Implementation
- 5 Results
- 6 Future Work
- 7 Conclusion

# Introduction

# The Clinical Burden

- Each endoscopic procedure requires a **detailed written report** — findings, diagnosis, and treatment recommendations
- Manual report writing takes **3–5 minutes per procedure**: a major bottleneck in high-volume GI units performing 30–50 cases per day
- A unit performing 40 procedures/day spends **~3.3 hours daily** on documentation alone — time taken directly away from patient care
- Repetitive documentation contributes to **physician burnout** and diagnostic fatigue  
Shanafelt et al. [1], Rotenstein et al. [2], National Academies of Sciences, Engineering, and Medicine [3]
- Documentation load has grown substantially with electronic health record requirements, doubling in the past decade National Academies of Sciences, Engineering, and Medicine [3]

**Bottom line:** Time spent writing reports is time *not* spent on patient care.

# The AI Gap

- General-purpose MLLMs (GPT-4V, LLaVA, Gemini) perform well on natural images but **fail on specialised endoscopy**:
  - Cannot reliably identify subtle mucosal features (erythema, granularity, friability)
  - Unfamiliar with endoscopic grading systems (Paris classification, Barrett's Prague criteria, ESGE grades)
  - Produce generic descriptions rather than **structured clinical reports**
- Existing medical MLLMs (LLaVA-Med, HuatuoGPT-Vision) are trained on broad biomedical data — **not** on physician-annotated endoscopic reports
- **No model existed** trained on GMC Calicut physician-annotated GI endoscopic data, leaving a critical gap for Indian clinical settings

## MVLMERG — Multimodal Vision-Language Model for Endoscopic Report Generation

Fine-tune a vision-language model to automatically generate professional-grade GI endoscopic reports.

- **Architecture:** LLaVA-4 base with CLIP-ViT-L/14 vision encoder + Vicuna-7B-v1.5 LLM backbone
- **Fine-tuning strategy:** P-LoRA — soft prompt tuning + selective LoRA
- **Two-stage training:**
  - ① Stage 1: Medical domain knowledge injection (MedTrinity-25M, ~160K Medical records)
  - ② Stage 2: Physician-style report generation (GMC Calicut, ~31,200 augmented samples)
- **Target:** Automated, structured, interpretable endoscopic reports in **~7 seconds** vs. 3–5 minutes manual

# Problem Definition

## Core Problem:

- Writing endoscopic image reports requires deep domain expertise in gastroenterological terminology, lesion morphology, and clinical logic
- General Multimodal Large Language Models (MLLMs) exhibit critical knowledge gaps: they struggle with specialised endoscopy terminology, pathological feature descriptions, and complex medical reasoning
- Existing fine-tuning methods suffer from **limited domain adaptability** and **insufficient endoscopic-specific knowledge**
- High-quality physician-annotated endoscopic data is scarce — publicly available medical datasets (PubMed) cover general medicine but lack the specificity needed for endoscopic report generation

**Goal:** Develop **MVLMERG** — a fine-tuned vision-language model that bridges the domain gap, producing professional-grade endoscopic reports matching the standard of intermediate-level physicians. Ali et al. [4], Haghighi et al. [5]

## Related Work: General Medical MLLMs

- **LLaVA-Med** Li et al. [6] — aligns PMC-15M biomedical image-text pairs in Stage 1; uses GPT-4-generated instruction data OpenAI [7] for end-to-end fine-tuning in Stage 2; strong generalization but **lacks endoscopy-specific terminology**
- **HuatuoGPT-Vision** Chen et al. [8] — refines medical image-text pairs from PubMed; MLLM-based denoising to build PubMedVision dataset; strong on broad medical VQA but **insufficient for detailed GI endoscopic findings**
- **BiomedGPT** Zhang et al. [9] — unified generative pre-trained transformer across vision, language, and multimodal biomedical tasks; **broad coverage but weak on GI endoscopic report generation**
- **PeFoMed** Haghighi et al. [5] — parameter-efficient fine-tuning framework for medical MLLMs; improves performance on medical imaging tasks but **not endoscopy-specific**

**Common gap:** Broad biomedical pre-training cannot substitute for **physician-annotated, domain-specific** endoscopic training data.



## Related Work: Surgical / GI-Specific Models

- **SurgicalGPT** — fine-tuned GPT for surgical visual Q&A; 3–5% accuracy gain over generalist models but focuses on **surgical scenes**, not endoscopic mucosal analysis
- **ColonGPT / ColonINST** — colonoscopy instruction tuning with polyp detection focus; good for classification but **not designed for full report generation**
- **EndoChat** — endoscopy dialogue system; limited to conversational Q&A format, **cannot produce structured clinical reports**

### What all GI-specific models lack:

- None trained on **physician-annotated, full-format** endoscopic reports
- None target **Indian clinical settings** (terminology, grading conventions differ)
- None support complete structured report generation (findings + diagnosis + recommendations)

## Related Work: Fine-Tuning Methods

- **LoRA** Hu et al. [10] — Low-Rank Adaptation; adds rank-decomposition matrices  $\Delta W = BA$  to frozen weights; highly parameter-efficient but **no mechanism to align input representations** with the target domain
- **Prefix / P-Tuning** Liu et al. [11] — soft prompt tokens prepended to input; aligns context but **without weight adaptation the model's knowledge remains unchanged**
- **PeFoMed** — parameter-efficient fine-tuning for medical LLMs; effective for general medical Q&A but **not evaluated on endoscopic scenarios**
- **MISS** — parameter-efficient medical MLLM fine-tuning; covers multiple imaging modalities but **not endoscopy-specific**

**Key insight:** Neither prompt tuning nor LoRA alone is sufficient — their **combination (P-LoRA)** addresses both input representation alignment *and* weight-space domain adaptation simultaneously.

## The Gap

No existing model combines (1) prompt-based input alignment, (2) selective weight adaptation, and (3) physician-annotated endoscopic training data.

- **P-LoRA** — unifies soft prompt tuning + selective LoRA: addresses both input alignment and domain knowledge injection in a single fine-tuning strategy
- **GMC Calicut physician-annotated data** — 6,500 real procedures annotated by senior endoscopists, augmented to  $\sim 31,200$  samples; reflects actual Indian clinical practice and terminology
- **Chain-of-Thought integration** — anatomical  $\rightarrow$  visual  $\rightarrow$  diagnostic reasoning baked into training data; produces interpretable, verifiable reports
- **First model** targeting professional Indian GI endoscopic report generation in a real hospital deployment setting

# Methodology

# Dataset Creation and Augmentation

- **Data Source:** Government Medical College (GMC), Calicut
- **Raw Data:** 6,500 **patient reports** (semi-structured PDFs + endoscopic images)
  - Each report includes **4 endoscopic images**
- **Data Augmentation:**
  - Applied **rotation and random translation** to every image
  - Purpose: increase dataset size and improve model robustness
  - **Approximate scale:**
    - Base Images = 9,000 (approx.)
    - Rotation alone multiplies by 4 → over 36,000 images
    - Translation variations further expand the pool
- **Text Data:**
  - Final image–report pairs after augmentation: ~**36,000 samples**
  - Data split: 34,150 training / 1,850 validation
- **Multimodal Association:** Each Q&A pair linked to original + augmented endoscopic images Wang et al. [12], Zhang and et al. [13]

# Figure 1: Custom Dataset Workflow

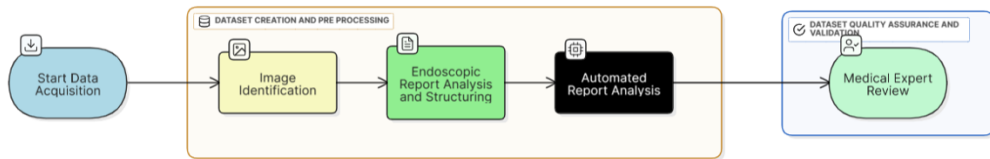


Figure: Data Workflow:  $\sim 36,000$  image–report pair generation Wang et al. [12], Zhang and et al. [13]

- Each original endoscopic image is rotated by **0°, 90°, 180°, and 270°**
- Produces exactly **4 variants** per image
- Clinically safe: tissue textures and lesion appearances remain unchanged
- **Effect:** 4× structured expansion of the base image set

- **Sampling:** Bernoulli process with  $p = 0.2$  selects approx. 20% of rotated images for translation
- **Shift model:** Translation vector  $(\tau_x, \tau_y)$  drawn from a truncated Gaussian:

$$\tau_x \sim \mathcal{N}(0, (0.05 W)^2), \text{ clamped to } [-0.15 W, +0.15 W]$$

$$\tau_y \sim \mathcal{N}(0, (0.05 H)^2), \text{ clamped to } [-0.15 H, +0.15 H]$$

- **Overlap guarantee:**

$$\text{Overlap Fraction} = \left(1 - \frac{|\tau_x|}{W}\right) \times \left(1 - \frac{|\tau_y|}{H}\right) \geq (1 - 0.15)^2 = 72.25\%$$

- **Result:** Controlled positional variation mimicking endoscopic camera jitter, without harming diagnostic content.



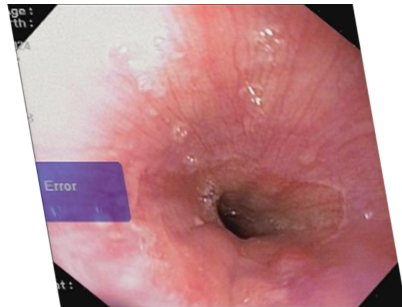
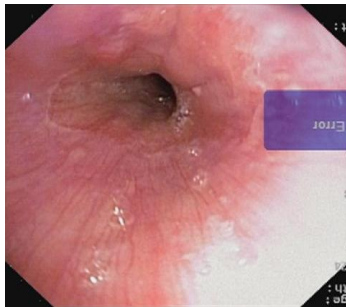
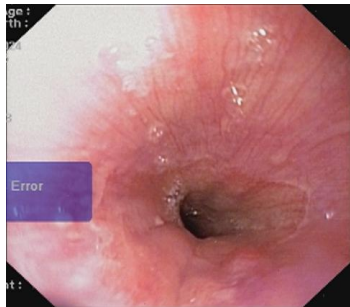


Figure: Left: Original    Middle: Rotated ( $90^\circ$ )    Right: Randomly translated

# Patient Data De-identification & Ethical Compliance

- **Regulatory Framework & Automated Script:**

- Strict adherence to the **Digital Personal Data Protection (DPDP) Act**, supported by a formal MoU between GMC Calicut and NIT Calicut
  - **Redaction approach mirrors rigorous de-identification used in recent high-profile document releases** – all personally identifiable fields irreversibly blacked out to ensure zero traceability
  - **Header Field Redaction:** Automatic blackout of Patient ID, Name, Age/Gender, Visit Date, UGI/OGD Numbers
- **Result:** Irreversible anonymization of all sensitive patient information before data enters the pipeline



## DEPARTMENT OF GASTROENTEROLOGY

MEDICAL COLLEGE HOSPITAL, KOZHIKODE  
[website; calicutmedicalcollege.ac.in](http://website.calicutmedicalcollege.ac.in)

Patient ID :  
Patient Name :  
Age/Gender :



Visit Date :  
Referred by :  
Performed by :



**Medical Entities Extracted:** Wang et al. [12], Zhang and et al. [13]

- Pathological findings and characteristics
- Diagnostic conclusions
- Procedural techniques and instruments

**Processing Method:**

- Natural language processing techniques
- Extraction of critical medical elements
- Foundation for Q&A pair generation

# Automated Question-Answer Pair Generation

**Multi-Stage Process:** Pogorelov and et al. [14]

- ① **Medical Entity Extraction** – Key concepts and diagnoses
- ② **Question Template Design** – Diagnostic, anatomical, pathological, procedural, comparative
- ③ **Automated QA Generation** – Multiple question variations per concept Li et al. [6]
- ④ **Answer Synthesis** – Accurate answers from reports + medical knowledge

**Result:** Diverse, clinically relevant question-answer combinations.

## Why GPT-5? OpenAI [7]

- Physician reports from GMC Calicut contain incomplete, colloquial, or inconsistent sentences
- GPT-5 acts as an **auxiliary annotation tool** to enrich and templatize raw descriptions
- Structured prompt: *“Convert to: This image shows (region). It describes (color, texture, features.)”*
- All GPT-5 outputs reviewed by senior gastroenterologists for clinical accuracy

## Semantic Consistency Filter: Reimers and Gurevych [15]

- Optimized sentences must preserve the original clinical meaning
- Sentence-BERT extracts vector embeddings  $\mathbf{u}$  (original) and  $\mathbf{v}$  (enhanced)
- Cosine similarity:  $S = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|}$  must satisfy  $S > 0.85$
- If  $S \leq 0.85$ , GPT-5 regenerates until threshold is met

# Dataflow Pipeline and Methodology

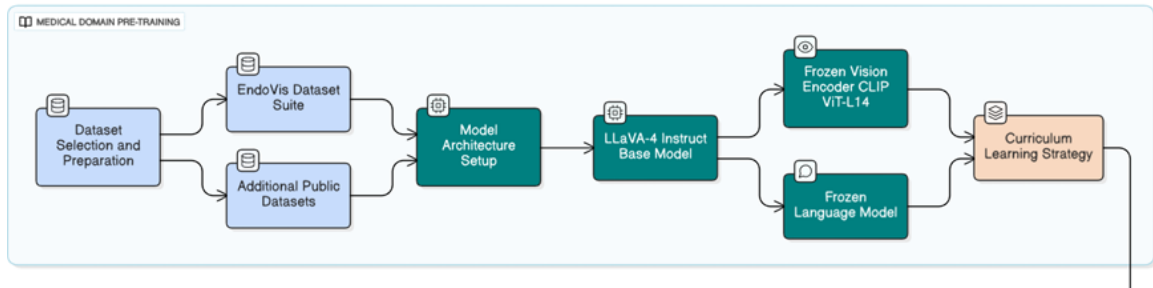


Figure: Stage 1: Pre-training.

# Dataflow Pipeline and Methodology

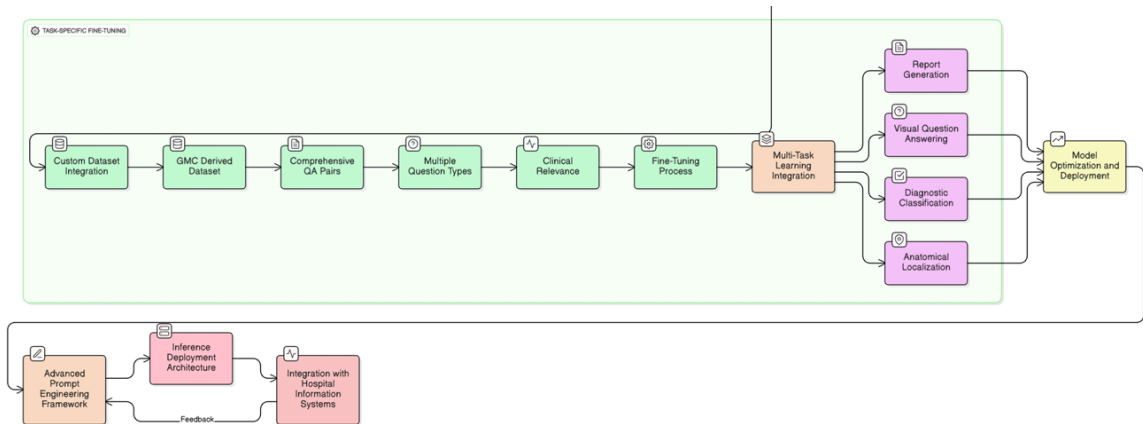


Figure: Stage 2: Fine-tuning

# Architecture



# Two-Stage Training Framework — MVLMERG

**Stage 1: Medical Domain Knowledge Injection** Leaman and Lu [16], Hu et al. [10], Zhang et al. [17]

- **Objective:** Acquire gastrointestinal domain knowledge
- **Base Model:** LLaVA-4 with frozen CLIP ViT-L/14
- **Trainable:** LLM backbone (vision encoder + connector frozen)
- **Datasets:** MedTrinity-25M ( $\sim 160\text{K}$  medical records) Xie et al. [18], Ben Abacha et al. [19]

**Stage 2: Professional Report Generation** Leaman and Lu [16], Hu et al. [10], Zhang et al. [17]

- **Objective:** Learn physician-style endoscopic report language
- **Dataset:** GMC Calicut physician-annotated data (6,500 base;  $\sim 31\text{K}$  after augmentation)
- **Strategy:** P-LoRA — prompt tokens + selective LoRA on key layers Hu et al. [10], Liu et al. [11], Zhang et al. [17]
- **Unfrozen:** LLM backbone (vision encoder + connector frozen)

P-LoRA combines **two complementary adaptation mechanisms**:

## Soft Prompt Tuning Liu et al. [11], Zhang et al. [17]

Prepends  $m = 64$  learnable tokens to the input embedding sequence — steers the model's *attention* toward endoscopic domain context before any weight computation.

- Prompt tokens correct **input representation**; LoRA adapters correct **weight-space knowledge**
- Only **0.343%** of all model parameters are trainable

## Selective LoRA Hu et al. [10], Zhang et al. [17]

Injects low-rank adapter matrices  $\Delta W = BA$  into the *first, middle, and last* transformer layers — adapts the model's *knowledge* to endoscopic terminology and clinical report style.

# Soft Prompt Tuning

**What it is:** Liu et al. [11]

- $m = 64$  learnable embedding vectors  $\{p_1, p_2, \dots, p_{64}\}$  prepended to the input token sequence
- Updated via gradient descent during training — **no manual prompt crafting required**
- At inference, the same learned tokens are prepended before every input

**Why we use it:**

- Manual prompts (e.g., “You are a gastroenterologist. . .”) are brittle and cannot encode fine-grained domain nuances
- Learnable tokens capture **implicit domain knowledge** — endoscopic terminology, physician report style, grading conventions — purely through the training signal
- The tokens guide the LLM’s attention toward the clinical domain *before* image and text tokens are processed
- Adds only  $64 \times 4096 \approx 262\text{K}$  parameters — negligible overhead with significant domain alignment benefit

# Selective LoRA

**What it is:** Hu et al. [10]

- Low-rank adapter matrices injected into weight matrices:  $W' = W + \Delta W = W + BA$
- $B \in \mathbb{R}^{d \times r}$ ,  $A \in \mathbb{R}^{r \times k}$ , rank  $r = 64$ , scaling  $\alpha = 32$ , dropout = 0.05
- Only  $B$  and  $A$  are trained; original  $W$  is frozen

**Why SELECTIVE (not all layers):**

- Transformer layers **specialise** at different levels of abstraction:
  - **Early layers** — low-level syntax and token patterns; minimal domain shift needed
  - **Middle layers** — semantic composition and concept associations; **critical for medical terminology**
  - **Last layers** — task-specific output formatting and clinical report style
- Targeting only these three groups **captures all domain adaptation needs** while training far fewer parameters
- Result: only **0.343%** trainable parameters vs.  $\sim 1\%$  for full LoRA — same quality, faster training, lower GPU memory

# Why P-LoRA Outperforms LoRA Alone

**Plain LoRA — the limitation:** Hu et al. [10], Liu et al. [11], Zhang et al. [17]

- Adapts weight matrices but does not change how the model *interprets the input context*
- For highly specialised domains like GI endoscopy, the model still “reads” input through a generic lens — producing suboptimal attention over clinical features

**P-LoRA — the combination advantage:**

- **Prompt tokens** shift input representations toward the endoscopic domain *before* attention is computed
- **LoRA adapters** then operate on already-domain-aligned representations — more efficient weight updates with less training signal wasted
- The two mechanisms are **orthogonal**: prompt tuning acts at the input embedding level; LoRA acts at the weight matrix level

**Empirical gains over plain LoRA:**

- **+5% CIDEr** and **+4.4% ROUGE-L** over plain LoRA
- GPU memory: ~43 GB (comparable to plain LoRA ~44 GB)
- Inference: ~7 s/report

# System Architecture

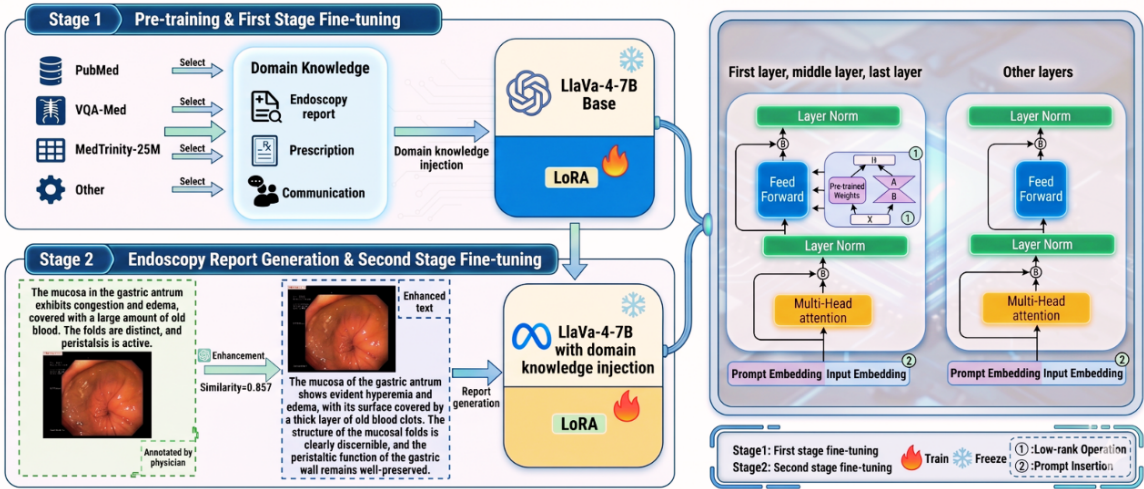
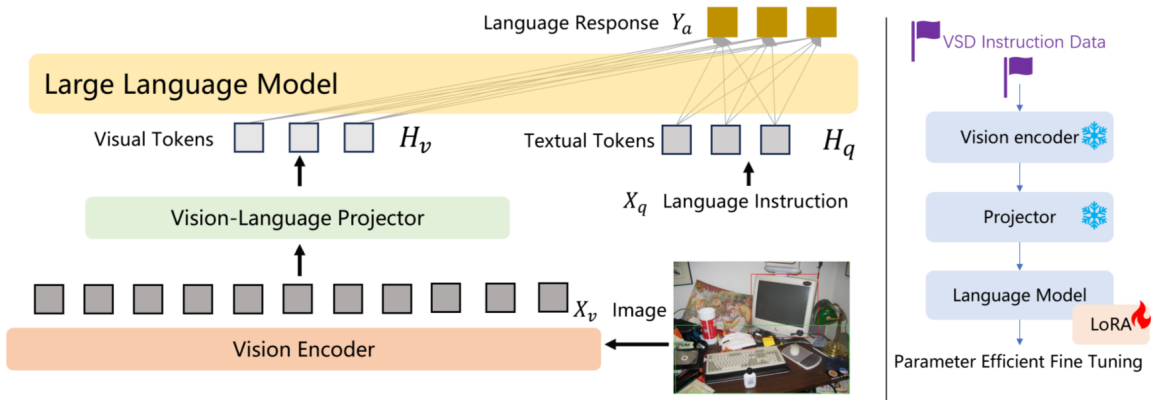


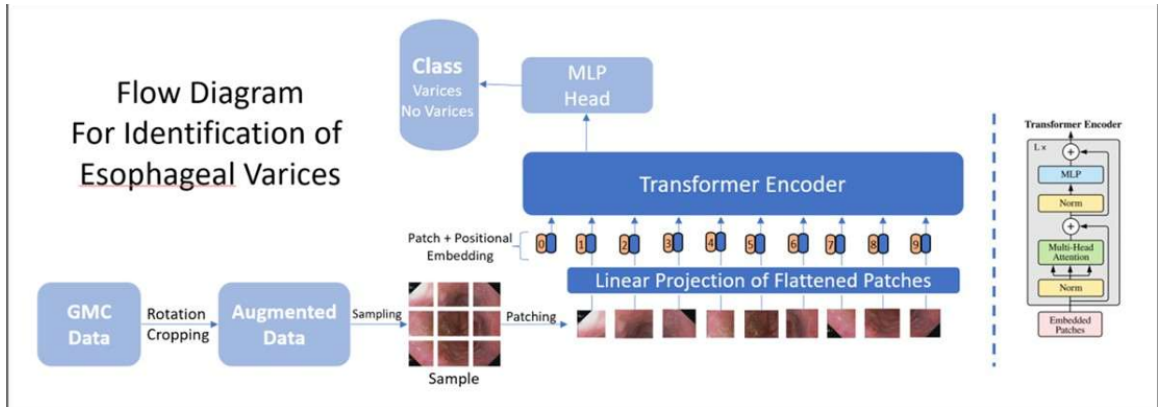
Figure: High-level architecture of the proposed vision-language model for endoscopic report generation

# Model Architecture – LLaVA



**Figure:** LLaVA architecture: vision encoder + projection layer + language model for multimodal understanding.

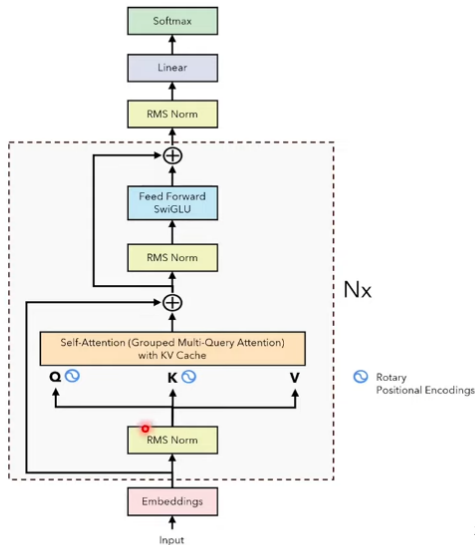
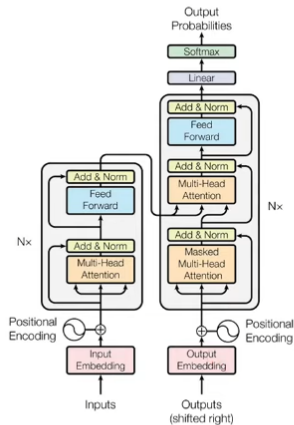
# CLIP-ViT Architecture



**Figure:** CLIP-ViT-L/14 vision encoder: patch embedding, transformer layers, and projected visual features.



## Transformer vs LLaMA



# Implementation

# Refining Generation: Implementing System Prompts

## Initial Challenge:

- Early model iterations produced verbose, unstructured, and redundant reports that did not match standard clinical formatting.

## Implemented Solution (Dataset Reformatting):

- Integrated rigorous **System Prompts** directly into the training dataset.
- **Role Assignment:** Instructed the model to assume a specific clinical persona (e.g., “You are an expert gastroenterologist...”).
- **Output Constraints:** Provided explicit instructions to prioritise direct, concise, and structured answers mirroring real GMC Calicut reports.

**Result:** A massive reduction in unnecessary text, yielding highly concise and structured outputs; further refinement improved diagnostic accuracy.

# Qualitative Impact: Verbose vs. Concise Reports

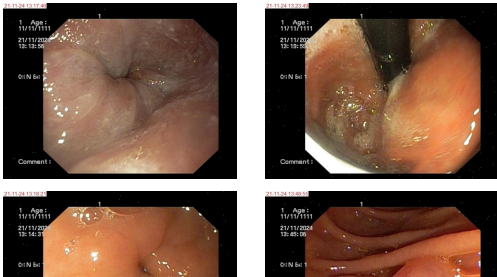
## Previous Generation (Verbose & Redundant)

### Medical Endoscopy Report

#### Patient Information

**Patient Name:** ABDUL MANAF  
**Patient ID:** ABDUL MANAF\_86376\_RP241121\_015818  
**Report Date:** 2025-12-04 12:55:59

#### Endoscopy Images



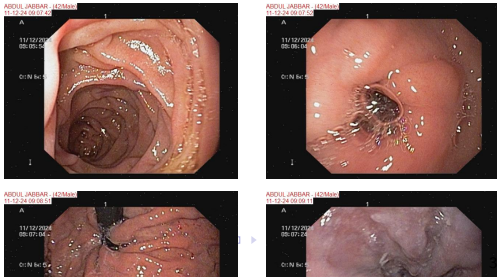
## Current Generation (Concise via System Prompt)

### Medical Endoscopy Report

#### Patient Information

**Patient Name:** ABDUL JABBAR  
**Patient ID:** ABDUL JABBAR\_84105\_RP241211\_091205  
**Report Date:** 2026-03-06 13:45:39

#### Endoscopy Images



# Qualitative Impact: Verbose vs. Concise Reports

## Previous Generation

*(Verbose & Redundant)*

|                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Scope Passed Upto: | D2 mucosa without ulcers/ erosions/ luminal narrowing .Scope passed upto D3 mucosa without any difficulty .Biopsy taken from D2 and D3 mucosa for HPR and CBNAAT from D2 and D3 mucosa . D1 mucosa was edematous and erythematous with superficial ulcers . D1 was not entered further due to poor patient tolerability . D1 mucosa was edematous and erythemat                                                                                                                |
| Premedication:     | 10% XYLOCAINE ANTRUM AND PYLORUS WITH SURROUNDING NODULARITY AND D1 NODULARITY FOR BIOPSY TAKEN RUT: PLAN: CECT THORAX + ABDOMEN AND BARIUM SWALLOW IF FURTHER EPISODE/HB FALL BIOPSY TAKEN FROM NODULARITY AND D1 NODULARITY FOR CBNAAT TAKEN RUT: PLAN: CECT THORAX + ABDOMEN AND BAR                                                                                                                                                                                        |
| Esophagus:         | Normal mucosa without ulcers/ erosions/ luminal narrowing with normal peristalsis and distensibility noted .No foreign body seen in the esophagus .No esophageal varices seen .No altered blood seen in the lumen .No luminal blood seen in the esophagus .No foreign body seen in the esophagus .No altered blood seen in the lumen .No luminal blood seen in the esophagus .No foreign body seen in the esophagus .No altered blood seen in the lumen .No luminal blood seen |
| Ge Junction:       | 40CMS and edematous and erythematous mucosa noted.Biopsy taken .No nodularity or ulcerations noted. No tear noted at present. No salivary pooling. No difficulty in swallowing. No foreign body seen. No luminal blood seen. No active ooze seen. No hemoptysis. No melena. No hematemesis. No hematuria. No proteinuria. No renal papilla. No ureter. No ampulla. No pyloric opening. No duodenal ulcer. No                                                                   |
| Stomach:           | No stasis . Normal peristalsis and distensibility noted . No ulcer/ erosions noted . Normal blood in lumen . No active bleed at present . Normal mucosal folds and contours noted . No luminal blood noted . Normal mucosal mucosa noted . No luminal blood noted . Normal mucosal mucosa noted . No luminal blood noted . Normal mucosal mucosa noted . No luminal blood noted . Normal mucosal                                                                               |
| Fundus:            | Normal mucosa without ulcers/ erosions/ luminal narrowing with normal distensibility and peristalsis noted .No food or fluid stasis noted .Normal distensibility and peristalsis noted .No food or fluid stasis noted .Normal distensibility and peristalsis noted .No food or fluid stasis noted .Normal distensibility and peristalsis noted .No food or fluid stasis noted .Normal                                                                                          |

## Current Generation

*(Concise via System Prompt)*

|             |                                                                  |
|-------------|------------------------------------------------------------------|
| Stomach:    |                                                                  |
| Fundus:     | No varices                                                       |
| Body:       | Mosaic pattern of mucosa seen                                    |
| Antrum:     | Normal                                                           |
| Pylorus:    | Normal                                                           |
| Duodenum:   | D1: Normal   D2: Normal                                          |
| Rut:        | Mosaic pattern of mucosa seen                                    |
| Biopsy:     | Not taken                                                        |
| Impression: | GRADE II X3 COLUMNS OF ESOPHAGEAL VARICES WITHOUT RCS SEVERE PHG |

Report generated by Medical LLaVA AI on 2026-03-06 at 13:45:39

# The “Blackbox” Problem & Our Solution

## The Initial Blackbox:

- The model would output a final diagnosis directly from an image without revealing its reasoning, making it hard to trust or debug.

## Our Solution – Chain-of-Thought Integration:

- Augmented the training Q&A pairs with **step-by-step reasoning chains**.
- Each chain forces the model to structure its answer as: **anatomical identification** → **visual observation** → **diagnosis**.
- During training, the model learns to mimic this transparent reasoning process.

**Outcome:** The model now explicitly explains *what it sees* and *why it concludes something*, turning a black-box predictor into an **interpretable diagnostic assistant**.

## Before CoT Implementation (Baseline Model):

- **Input:** [Image] “What is the diagnosis?”
- **Output:** “Grade II Esophageal Varices.” (*correct but no justification*)

## After CoT Implementation (Our Model):

- **Input:** [Image] “Analyze the image step-by-step and provide a diagnosis.”
- **Output:**
  - ➊ **Observation 1:** I observe the lower third of the oesophagus.
  - ➋ **Observation 2:** There are swollen, tortuous submucosal veins present.
  - ➌ **Observation 3:** The veins occupy less than 1/3 of the lumen, with no red colour signs.
  - ➍ **Conclusion:** Therefore, the diagnosis is Grade II Esophageal Varices.

**Impact:** Explicit reasoning enhances clinical trust, allows verification, and helps identify subtle mistakes – a crucial feature for real-world deployment.

# Fine-Tuning Configuration Details

## Vision Encoder: CLIP-ViT-L/14

- Image resolution:  $336 \times 336$  pixels
- Patch size:  $14 \times 14$  ( $24 \times 24$  grid)
- Total vision tokens: 577 per image (576 patches + CLS)

## Training Hyperparameters:

- **Precision:** bfloat16 (BF16)
- **Optimizer:** AdamW with 8-bit quantization
- **Learning Rate:**  $2e^{-4}$  with cosine scheduler, warmup 0.1
- **Batch Size:** 2 micro-batch, 4 accumulation steps = 8 global
- **Attention:** Flash Attention 2
- **Epochs:** 4 epochs (both stages)



# Training Configuration Rationale

- **CLIP-ViT-L/14** – pre-trained on 400M pairs; robust medical visual features; L size balances capacity vs. speed;  $14 \times 14$  patch grid captures both anatomy and small lesions.
- **$336 \times 336$  resolution** – standard for LLaVA; sufficient detail without excessive memory.
- **577 vision tokens** – each patch token carries localised information, enabling the language model to “see” global and local structures.
- **bfloat16** – halves memory vs. FP32, no performance loss.
- **AdamW + 8-bit** – stable weight-decoupled convergence; 8-bit reduces optimizer memory 75%.
- **LR  $2e^{-4}$ , cosine + warmup** – low peak prevents catastrophic forgetting; warmup stabilises early steps.
- **Batch size 8 (accumulated)** – simulates larger batch under GPU memory limits.
- **Flash Attention 2** – memory-efficient exact attention; 5-10× VRAM reduction for long sequences.
- **4 epochs** – sufficient for P-LoRA adapters on the augmented second-stage dataset without overfitting.

# Why AdamW & Flash Attention 2?

## AdamW over Adam

- Adam couples weight decay with gradient scaling – can over-regularize.
- AdamW **decouples** weight decay from adaptive gradients → better generalization.
- Standard for LLaMA fine-tuning; prevents over-shrinking of LoRA adapters.

## Flash Attention 2 over regular / Flash Attention 1

- Standard attention:  $O(N^2)$  memory – prohibitive for 577 vision tokens + text.
- Flash Attention 1 tiled computation, but under-utilised GPU parallelism.
- Flash Attention 2:
  - **Fewer non-matmul operations** → higher throughput.
  - **Better batch/head parallelism** → faster training.
  - **Lower VRAM usage (5-10× reduction)** → feasible on single GPU.

**Together:** Stable convergence + memory-efficient training at scale.

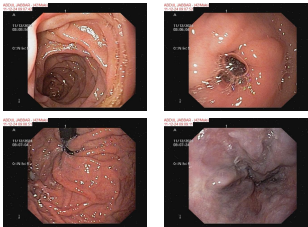
# Results

## Medical Endoscopy Report

### Patient Information

**Patient Name:** ABDUL JABBAR  
**Patient ID:** ABDUL JABBAR\_84105\_RP241211\_091205  
**Report Date:** 2026-04-13 18:09:26

### Endoscopy Images



### Findings

|                           |                  |
|---------------------------|------------------|
| <b>Indication:</b>        | CLD-UGIB         |
| <b>Scope Passed Upto:</b> | D2               |
| <b>Premedication:</b>     | Lignocaine spray |
| <b>Esophagus:</b>         | Normal           |
| <b>Ge Junction:</b>       | Lax at 38cm      |

|                    |                                                          |
|--------------------|----------------------------------------------------------|
| <b>Stomach:</b>    |                                                          |
| <b>Fundus:</b>     | Edematous mucosa noted                                   |
| <b>Body:</b>       | Edematous mucosa noted                                   |
| <b>Antrum:</b>     | Erythematous mucosa noted                                |
| <b>Pylorus:</b>    | Normal                                                   |
| <b>Duodenum:</b>   | D1: Normal   D2: Normal                                  |
| <b>Rut:</b>        | Not taken                                                |
| <b>Biopsy:</b>     | Not taken                                                |
| <b>Impression:</b> | GRADE II ESOPHAGEAL VARICES WITH RCS EVL DONE SEVERE PHG |

Report generated by Medical LLaVA Ai (FTT) on 2026-04-13 at 18:09:26

# Evaluation Metrics

## CIDEr

Consensus-based evaluation using TF-IDF weighted n-gram vectors.

$$\text{CIDEr}(c, S) = \frac{1}{M} \sum_m \frac{\mathbf{g}^n(c) \cdot \mathbf{g}^n(s_m)}{\|\mathbf{g}^n(c)\| \|\mathbf{g}^n(s_m)\|}$$

## ROUGE-L

F-measure based on longest common subsequence (LCS).

$$\text{ROUGE-L} = \frac{(1 + \beta^2) R_{lcs} \cdot P_{lcs}}{R_{lcs} + \beta^2 P_{lcs}}$$

## METEOR

Combines precision, recall, and a fragmentation penalty.

$$F_{mean} = \frac{10PR}{R + 9P}, \quad \text{METEOR} = F_{mean}(1 - \text{Pen})$$

## BLEU-4

Geometric mean of 1–4 gram precisions with brevity penalty.

$$\text{BLEU-4} = \text{BP} \cdot \exp\left(\frac{1}{4} \sum_{n=1}^4 \log p_n\right)$$

## BERTScore

Semantic similarity via contextual BERT token embeddings.

$$F_{\text{BERT}} = \frac{2 P_{\text{BERT}} \cdot R_{\text{BERT}}}{P_{\text{BERT}} + R_{\text{BERT}}}$$

All five metrics reward similarity to reference reports; higher is better ( $\uparrow$ ).

# Why These Evaluation Metrics?

## **CIDEr** — *primary metric*

- Designed for report generation; TF-IDF up-weights *discriminative* n-grams, penalising generic filler
- Best reflects domain-specific clinical term quality

## **ROUGE-L** — *clinical structure*

- LCS-based: captures correct finding sequence (anatomy → observation → diagnosis)
- Robust to paraphrasing

## **METEOR** — *linguistic fluency*

- Handles synonyms and stemming — physicians use varied clinical vocabulary
- Fragmentation penalty discourages

## **BLEU-4** — *n-gram precision*

- 4-gram precision captures multi-word phrases (e.g., “Grade II esophageal varices”)
- Included for comparability with prior medical NLG literature

## **BERTScore** — *semantic correctness*

- Contextual BERT embeddings measure semantic similarity beyond surface overlap
- Handles spelling variants (“oesophageal” vs. “esophageal”)
- Strongest indicator of correct clinical meaning

## Quantitative Comparison — MVLMERG vs. Baselines

| Method               | CIDEr $\uparrow$ | ROUGE-L $\uparrow$ | METEOR $\uparrow$ | BLEU-4 $\uparrow$ | BERTScore $\uparrow$ |
|----------------------|------------------|--------------------|-------------------|-------------------|----------------------|
| Qwen2-VL-7B-Instruct | 0.0078           | 0.0504             | 0.0564            | 0.0009            | 0.8317               |
| BLIP-2               | 0.0079           | 0.0147             | 0.0347            | 0.0002            | 0.7957               |
| InstructBLIP         | 0.0088           | 0.0320             | 0.0583            | 0.0004            | 0.8196               |
| <b>Our Model</b>     | <b>0.0531</b>    | <b>0.2632</b>      | <b>0.1187</b>     | <b>0.0006</b>     | <b>0.8760</b>        |

Evaluated on 100 samples. MVLMERG leads on 4 of 5 metrics (CIDEr, ROUGE-L, METEOR, BERTScore).

*Note: BLEU-4 (0.0006) is marginally below Qwen2-VL (0.0009); all other metrics show clear improvement.*

## Evaluation Protocol: Ali et al. [4]

- 2 senior endoscopists (10+ years experience), GMC Calicut
- Each generated report labelled **Correct** / **Incorrect** based on clinical accuracy
- Disagreements resolved by **joint review and consensus discussion**



# Hardware & Training Setup

## Training Hardware:

- **GPU:** NVIDIA H200 NVL 143 GB HBM3e (CUDA 13.0)
- **CPU:** Intel Xeon 6505P (12 cores/socket, 2 sockets, 48 threads)
- **RAM:** 251 GiB
- **Python:** 3.12.3; PyTorch 2.11.0+cu130 (BF16 + Flash Attention 2)

## Training Resource Usage:

- GPU memory during training:  $\sim 43$  GB
- Trainable parameters: **0.343%** of full model

## P-LoRA Hyperparameters:

- LoRA rank  $r = 64$ ,  $\alpha = 32$ , dropout = 0.05
- Prompt tokens = 64
- Epochs: 4 (both stages)
- Seed: 42
- Max sequence length: 4096

## Inference:

- Average latency:  $\sim 7$  s per report

# Future Work

# Model Optimization & Quantization

## GGUF Format Conversion Dettmers et al. [20], Frantar et al. [21]

- Optimizes CPU and mixed CPU-GPU inference
- Significantly improved model loading times Frantar et al. [21]

## Multi-Level Quantization Strategy Dettmers et al. [20], Frantar et al. [21]

| Format | Use Case                    | Size Reduction                     |
|--------|-----------------------------|------------------------------------|
| Q8_0   | High-end workstations       | ~50%                               |
| Q5_K_M | Standard clinical computers | ~69%                               |
| Q4_K_M | Basic clinical terminals    | ~75% (3.25GB) Dettmers et al. [20] |
| Q3_K_L | Mobile deployments          |                                    |
|        |                             | Ultra-compact                      |

**Site-Specific Adaptation:** Brief P-LoRA fine-tuning at each new hospital site to adapt to local reporting vocabulary and endoscope characteristics — measuring adaptation cost across  $\geq 2$  hospitals.

# Deployment Planning

## Two Deployment Strategies:

- ① **vLLM Framework** Kwon et al. [22]
  - High-throughput clinical environments
  - Concurrent access scenarios
- ② **Ollama Framework** Kwon et al. [22]
  - Individual workstations
  - Low-resource environments
  - Resource-constrained deployment

## EHR Integration:

- RESTful API architecture (secure, authenticated endpoints)
- Real-time synchronization with patient records

## Regulatory Pathway:

- Pending IEC approval and **CDSCO SaMD** (Software as a Medical Device) compliance before live clinical use

# Limitations

- **Single-institution data:** Trained on GMC Calicut data only; performance may vary across hospitals with different endoscope models, lighting, and reporting vocabulary.
- **Single-frame input:** Processes individual frames in isolation; clinicians review multiple sequential images before reporting — temporal context is lost.
- **No uncertainty quantification:** Model does not communicate its own confidence level; rare but clinically relevant errors do occur.
- **Under-represented categories:** Early Barrett's oesophagus, post-ESD healing patterns, and rare lesion types are sparse in the training set.
- **Mandatory physician review:** All generated reports must be reviewed and approved by a qualified endoscopist before influencing any patient care decision.

# Conclusion

## Key Achievements:

- Built a vision-language model for automated endoscopic report generation, fine-tuned on a custom  $\sim 31,200$ -sample dataset.
- Implemented a DPDP-compliant de-identification pipeline, making sensitive clinical data safe for AI training.
- Integrated system prompts and Chain-of-Thought reasoning to produce **concise, transparent, and clinically interpretable** reports.
- Achieved state-of-the-art on 4/5 metrics — CIDEr **0.0531**, ROUGE-L **0.2632**, METEOR **0.1187**, BERTScore **0.8760**; BLEU-4 0.0006 (slightly below Qwen baseline 0.0009).

**Impact:** Drastically reduces documentation time; provides explainable AI assistance to gastroenterologists Shanafelt et al. [1], Rotenstein et al. [2]. **Path Forward:** Model quantization

Dettmers et al. [20], Frantar et al. [21] and clinical deployment trials (pending IEC approval) will bring the tool into real-world endoscopic workflows.

# Acknowledgments

- **Government Medical College (GMC), Calicut**
  - Provided the dataset of 6,500 procedures and invaluable clinical expertise.
- **Medical Staff**
  - Continuous feedback and validation throughout development.
- **Mentors & Colleagues**
  - Project guidance and support from NIT Calicut faculty.

**Thank You!**

# References I

- [1] Tait D. Shanafelt, Colin P. West, Lotte N. Dyrbye, Hanhan Wang, Christine Sinsky, and Mickey Trockel. Changes in burnout and satisfaction with work-life integration in physicians and the general us working population between 2011 and 2020. *Mayo Clinic Proceedings*, 97(3):491–506, 2022. doi: 10.1016/j.mayocp.2021.11.021.
- [2] Lisa S. Rotenstein, Melisa Abola, Sachin H. Jain, and Christine K. Cassel. Clinician burden and the impact on patient care: a qualitative study. *BMJ Open Quality*, 12(2): e002084, 2023. doi: 10.1136/bmjopen-2022-002084.
- [3] National Academies of Sciences, Engineering, and Medicine. *Reducing Administrative Burden in Health Care: Strategies to Improve Clinician Well-Being*. National Academies Press, 2024.
- [4] Heya Ali, Muhammad A. Muzammil, Dushyant S. Dahiya, Fawad Ali, Shanza Yasin, Warda Hanif, Madhav K. Gangwani, Muhammad Aziz, Moataz Khalaf, Debashis Basuli, and Mohammad Al-Haddad. Artificial intelligence in gastrointestinal endoscopy: A comprehensive review. *Annals of Gastroenterology*, 2024.



## References II

- [5] Fatemeh Haghighi, Mohammad Reza Hosseinzadeh Taher, Zongwei Zhou, Michael B. Gotway, and Jianming Liang. Medical vision-language models: A survey. *Frontiers in Artificial Intelligence*, 2024.
- [6] Chunyuan Li, Cliff Wong, Steven Zhang, Naoto Usuyama, Haotian Liu, Jianwei Yang, Tristan Naumann, Hoifung Poon, and Jianfeng Gao. Llava-med: Training a large language and vision assistant for biomedicine. In *NeurIPS*, 2023.
- [7] OpenAI. GPT-4 technical report. *arXiv preprint arXiv:2303.08774*, 2023.
- [8] Jiarong Chen, Junying Li, Weichen Wang, Zhenyang Liu, Guiming Hardy Li, et al. Huatuoqpt-vision, towards injecting medical visual knowledge into multimodal LLMs at scale. *arXiv preprint arXiv:2406.19280*, 2024.
- [9] Kai Zhang, Jun Yu, Zhiling Yan, Yixin Liu, Eashan Adhikarla, Sunyang Fu, Ying Chen, Yuyin Zhou, Xiang Li, Lifang Jiang, et al. Biomedgpt: A unified and generalist biomedical generative pre-trained transformer for vision, language, and multimodal tasks. *arXiv preprint arXiv:2305.17100*, 2023.

## References III

- [10] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. Lora: Low-rank adaptation of large language models. In *International Conference on Learning Representations (ICLR)*, 2022.
- [11] Xiao Liu, Kaixuan Ji, Yicheng Fu, Weng Lam Tam, Zhengxiao Du, Zhilin Yang, and Jie Tang. P-tuning v2: Prompt tuning can be comparable to fine-tuning universally across scales and tasks. In *Proceedings of the ACL*, 2022.
- [12] Xiaosong Wang, Yifan Peng, Le Lu, Zhiyong Lu, and Ronald M. Summers. Automated medical qa generation: A review. *arXiv preprint*, 2024.
- [13] Yuhao Zhang and et al. Medical instruction datasets for large language models. *JMIR*, 2024.
- [14] Konstantin Pogorelov and et al. Kvasir-vqa-x1: Medical visual question answering dataset. *arXiv preprint*, 2024.

- [15] Nils Reimers and Iryna Gurevych. Sentence-BERT: Sentence embeddings using Siamese BERT-networks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing*, pages 3982–3992. Association for Computational Linguistics, 2019. doi: 10.18653/v1/D19-1410.
- [16] Robert Leaman and Zhiyong Lu. Clinical applications of large language models. *ACL Anthology*, 2024.
- [17] Xingyu Zhang, Xiangwei Zheng, Zhen Li, Mingjun Ma, Xiao Liang, Xiuxiu Ren, and Mingzhe Zhang. A fine-tuning multimodal large language model for endoscopic report generation. *Biomedical Signal Processing and Control*, 118:109737, 2026. doi: 10.1016/j.bspc.2026.109737.
- [18] Yunfei Xie, Ce Wu, Pengfei He, et al. Medtrinity-25m: A large-scale multimodal dataset with multigranular annotations for medicine. In *International Conference on Learning Representations (ICLR)*, 2025.

- [19] Asma Ben Abacha, Sadid A. Hasan, Vivek V. Datla, Joey Liu, Dina Demner-Fushman, and Henning Müller. Vqa-med: Overview of the medical visual question answering task at imageclef 2019. In *CLEF 2019 Working Notes*, 2019.
- [20] Tim Dettmers, Artidoro Pagnoni, Ari Holtzman, and Luke Zettlemoyer. QLoRA: Efficient finetuning of quantized LLMs. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
- [21] Elias Frantar, Saleh Ashkboos, Torsten Hoefler, and Dan Alistarh. GPTQ: Accurate post-training quantization for generative pre-trained transformers. In *International Conference on Learning Representations (ICLR)*, 2023.
- [22] Woosuk Kwon, Zhuohan Li, Siyuan Zhuang, Ying Sheng, Lianmin Zheng, Cody Hao Yu, Joseph E. Gonzalez, Hao Zhang, and Ion Stoica. Efficient memory management for large language model serving with PagedAttention. In *Proceedings of the 29th ACM Symposium on Operating Systems Principles (SOSP)*, 2023.